

Conformal Calibration of Open-Vocabulary Semantic Maps under Label and Pose Uncertainty

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Abstract—A robot that avoids “the plant” on an open-vocabulary semantic map receives two errors at once: the detector under-covers or mislabels the object, and the pose estimate drifts, so the object is drawn in the wrong place. Planners with distribution-free guarantees calibrate the perception error and assume a known pose. We calibrate a single per-scene quantity, the *miss distance*: how far the true footprint of an object sticks out of its class region on the map built with the estimated poses, measured in the planner’s frame. Its split-conformal quantile is a keep-out margin that absorbs both errors and certifies any planner. On 21 OSMa-Bench ReplicaCAD scenes with YOLO-World, the joint margin holds 0.93–0.94 coverage at a 0.9 target under drift from 0 to 45 cm median ATE, while margins calibrated on labels or on pose alone fall to 0.76 and ≤ 0.22 , and per-cell label sets fall to 0.73 at 2 cm. In the setting of the closest prior work (known pose, closed vocabulary) per-cell label calibration and ours achieve the same mission success (0.43 vs. 0.42); with drift its coverage drops to 0.10 while ours stays at 0.94, and with an open vocabulary ours succeeds 1.8 $\times$ more often. We also show that the sum of separately calibrated margins can under-cover, that real visual odometry without loop closure leaves no useful certificate, and that the remaining price of the guarantee is set by the worst-case perception error, not by the risk level.

I. INTRODUCTION

Open-vocabulary semantic maps [1] let a robot follow instructions such as “keep away from the bike”. Such a map is assembled from two unreliable sources. Labels come from a detector or segmenter that misses, truncates and mislabels objects; geometry comes from a pose estimate that drifts. Benchmarks quantify how strongly these maps degrade, e.g. OSMa-Bench [2] reports the f-mIoU of ConceptGraphs dropping from 28.00 to 14.29 under changed lighting, but they score the map as an image, without a robot moving through it. A recent survey of 3D scene graphs [4] names the missing link between map errors and robot behaviour as an open problem.

Conformal prediction (CP) [5] gives planners distribution-free guarantees on top of learned perception. Sundarsingh et al. [6] calibrate per-cell label sets of a semantic grid map and plan with the most conservative label; Perceive With Confidence [7] calibrates the inflation of an occupancy map. Both assume that the robot’s pose is known. Pose error, however, is real and poorly calibrated in semantic memory: on a real robot, loop closures shift keyframes by meters and the nominal 95 % pose ellipsoid covers only a third of the

actual errors [8]. Separately calibrated modules also need not compose into a calibrated system [9].

We ask what guarantee a planner gets when label error and pose error act together, and how it should be calibrated. Our contributions are:

- a nonconformity score, the *miss distance*, measured in the planner frame on a map built with estimated poses, whose per-class conformal quantile is a keep-out margin covering both error sources for any planner;
- a controlled study on 21 OSMa-Bench scenes with six drift levels and real visual odometry, showing that margins calibrated on one error source lose coverage under the other, and that the sum of separately calibrated margins over-pays or under-covers depending on the unknown dependence of the errors;
- a comparison with per-cell label-set calibration [6] in its own setting and in ours, which explains why label-space calibration degenerates on indoor maps;
- an account of the price of the guarantee in navigation tasks and of where it comes from.

II. RELATED WORK

Open-vocabulary semantic mapping. ConceptGraphs [1] fuses class-agnostic masks into an object graph with CLIP features; OSMa-Bench [2] and OSMa-Bench++ [3] evaluate such mappers under lighting changes. Naive Bayesian fusion makes semantic maps over-confident [10]; we fuse by averaging. None of these works evaluates the map through a motion task.

Conformal guarantees for planning. CP was introduced to planning through trajectory prediction of other agents [11]. Closest to us, Sundarsingh et al. [6] calibrate label sets of a semantic grid with one score per environment (the worst cell) and report 93.4 % map and 98.4 % mission success at a 95 % target, with a known pose, five classes and a 1 m grid in 70×25 m outdoor scenes. Perceive With Confidence [7] calibrates the inflation of an occupancy map that makes it contain all obstacles; our score is a semantic analogue of theirs, with pose error included. Learnable CP [12] learns a context-aware score for RRT* clearance and includes synthetic localization noise; Conformal Constraint Tightening [13] shows that tightened constraints are safe for any planner and that sampling-based planners solve fewer tightened problems.

Composition and pose error. Baral [9] shows that marginally calibrated modules give system-level coverage anywhere between 86 % and 99.98 % depending on error corre-

lation and that joint calibration restores it. P-POSEMEM [8] documents how pose-graph rewrites displace objects in semantic memory. To our knowledge, no prior work calibrates open-vocabulary label error and pose error jointly with a planner in the loop; this claim rests on a citation and keyword search of October 2026 and is preliminary.

III. PROBLEM FORMULATION

A robot traverses a scene Ω with true camera poses $T_k \in SE(3)$ and estimated poses $\hat{T}_k = D_k T_k$, where D_k is the accumulated error. A top-down grid map $\hat{M}$ is fused from all frames placed with $\hat{T}_k$; a cell j stores the score-weighted fraction $p_j(k)$ of its observations that support class k . The region of class k at threshold λ is

$$A_\lambda(k) = \{j : j \text{ occupied}, p_j(k) \geq \lambda\}. \quad (1)$$

The planner acts in the frame the robot believes in at query time q . An object seen at frame k is drawn at $D_k p$ but is truly at $D_q p$, so in the planner frame the true footprint of object e is $\hat{F}(e) = D_q F(e)$: the map is wrong both because of the label and because of the drift accumulated between observation and query. The robot must reach a goal while keeping a distance d from every object of the avoid classes (here $d = 0.5$ m).

IV. METHOD

Miss distance. For a scene Ω and class k ,

$$s_k(\Omega) = \min \left\{ R_{\max}, \max_{e: k_e=k} \max_{x \in \hat{F}(e)} \text{dist}(x, A_\lambda(k)) \right\}. \quad (2)$$

A missed or mislabeled object makes s_k large; a pose shift makes it equal to the shift. One number absorbs both errors. Scores are computed once per scene: cells, frames and objects of a scene are strongly correlated and are not exchangeable units.

Calibration. Given n calibration scenes, $\hat{q}_k$ is the $[(n+1)(1-\alpha)]$ -th smallest score. If $\hat{q}_k \geq R_{\max}$ or $n < 1/\alpha - 1$, the class *abstains*: its certificate is vacuous. Calibration is per class (Mondrian), so one unreliable class does not inflate the others.

Proposition 1. *If the calibration scenes and the test scene are exchangeable, then with probability at least $1 - \alpha$ every true footprint of class k lies within $\hat{q}_k$ of $A_\lambda(k)$. Any path that keeps $\text{dist}(x, A_\lambda(k)) \geq d + \hat{q}_k$ keeps a true distance of at least d to every object of class k with the same probability.*

Proof. The first claim is split conformal prediction [5] applied to s_k . If $s_k \leq \hat{q}_k$, every point y of a true footprint has a point $a \in A_\lambda(k)$ with $\|y - a\| \leq \hat{q}_k$; for a path point x , $\|x - y\| \geq \|x - a\| - \|a - y\| \geq d + \hat{q}_k - \hat{q}_k = d$. $\square$

Compared calibrations. All arms share the region $A_{\lambda_0}(k)$ ($\lambda_0 = 0.05$, chosen on a development scene) and differ in what the margin is calibrated on: *uncalibrated* (margin 0); *labels only* (detector map at true poses); *pose only* (map with true labels and drifted poses); *separate* (sum of the two, each at level α); *joint* (detector map with drifted poses, Eq. (2)). We also re-implement *per-cell label sets* [6]: the score is the worst

$1 - p_j(k)$ over true footprint cells, the threshold is $\lambda^* = 1 - \hat{q}$, and the planner avoids every cell whose label set contains k .

Why the sum of quantiles is not enough. If the joint miss is at most the label miss plus the pose displacement, the union bound only gives $P(\text{both within their quantiles}) \geq 1 - 2\alpha$. Bonferroni ($\alpha/2$ each) restores $1 - \alpha$ but needs $n \geq 2/\alpha - 1$ scenes (19 at $\alpha = 0.1$). Where the sum lands between $1 - 2\alpha$ and 1 depends on the dependence of the errors, which is unknown in practice; the joint quantile does not depend on it.

V. EXPERIMENTAL SETUP

Data and perception. 22 ReplicaCAD scenes of OSMA-Bench [2], [14] (RGB-D, ground-truth semantics and poses, 640×480, every 8th frame). Scene apt_0 is used for all design choices; the other 21 scenes for calibration and test. YOLO-World v2-s [15] with the 98 object classes of the dataset as prompts; inside each box, pixels within 0.25 m of the box's median depth are kept. Maps are 5 cm grids; occupancy is taken from points 0.1–1.8 m above the floor. Avoid classes: indoor plant, bike, TV stand.

Pose error. Synthetic drift: every relative motion of the true trajectory is replayed with planar noise proportional to the distance travelled and the angle turned. Six levels are set once on the development scene by target absolute trajectory error (ATE) and applied unchanged, since the noise is per metre and per degree; on the 21 evaluation scenes, whose paths are shorter, the median ATE is L1 0.4 cm, L2 2 cm, L3 7 cm, L4 45 cm (3.6% of path) and L5 2.7 m. We label levels by these medians; 10 realizations per level. Real pose error: frame-to-frame RGB-D odometry of Open3D [16], [17] on every 2nd frame with a planar ground-robot constraint, no loop closure.

Calibration protocol. $\alpha = 0.1$ unless stated; 200 random splits into 13 calibration and 8 test scenes; each calibration scene contributes one random drift realization, test scenes all of theirs; $R_{\max} = 3$ m. Abstention counts as (vacuous) coverage and is reported separately.

Missions. Start and goal are ≥ 1.5 m from avoid objects and ≥ 3 m apart. *Pass-by* tasks are selected so that the shortest obstacle-free path passes within 0.5 m of an avoid object (233 tasks in 15 scenes); *random* tasks are not selected (20 per scene). Dijkstra plans on the predicted map with keep-out zones $A_\lambda(k) \oplus (d + \hat{q}_k)$; margins are calibrated leave-one-scene-out. A path is *unsafe* if it passes closer than d to a true avoid object in the planner frame; a mission *succeeds* if a safe path to the goal is found. A class that abstains falls back to treating every occupied cell as that class.

VI. RESULTS

A. Coverage under pose drift

Table I summarizes the coverage. The uncalibrated map almost never contains the full true footprint: detectors truncate objects at their edges. Per-cell label sets reach 1.00 without drift only through a threshold $\lambda^* = 0$, i.e. by declaring every occupied cell possibly hazardous, and they lose coverage at

TABLE I
COVERAGE OF THE GUARANTEE AND MEDIAN MARGIN (M) FOR THE BIKE AND THE PLANT, $1 - \alpha = 0.9$. [†] THE ARM ABSTAINS IN $\geq$ HALF OF THE SPLITS (MARGIN OF THE OTHERS SHOWN).

calibration	bike				indoor plant			
	L0	L2	L4	$\hat{q}$ L4	L0	L2	L4	$\hat{q}$ L4
uncalibrated	0.00	0.00	0.01	0	0.00	0.00	0.00	0
per-cell labels [6]	1.00	0.73	0.17	—	1.00	0.41	0.03	—
labels only	0.93	0.94	0.76	0.75	0.93	0.96	0.86	0.85 [†]
pose only	0.00	0.03	0.82	0.90	0.00	0.00	0.86	1.38
separate (sum)	0.93	0.98	0.97	1.60	0.93	0.98	0.95	2.10 [†]
joint (ours)	0.93	0.94	0.93	1.17	0.93	0.93	0.93	1.52[†]

2 cm of drift because they have no geometric slack. A labels-only margin holds while the drift is small compared with the label error (≤ 7 cm ATE) and fails at odometry-level drift. A pose-only margin cannot see detector errors. The joint margin holds 0.93–0.94 at every level and is 27% smaller than the sum of separate quantiles at L4. The plant abstains in about 0.6 of the splits because in one scene the detector misses it completely; the TV stand, labelled “bench” or “rug” in 2 of 21 scenes, abstains in 0.71–0.95 of the splits. With 13 calibration scenes the quantile is the maximum score, so a single such scene in the calibration set leaves the class uncertifiable — the correct answer for this detector.

B. Missions

On pass-by tasks the uncalibrated planner finds a path almost always, but 27–37% of its paths are unsafe at L0–L3. Every calibrated arm has no unsafe path at any level; the price is that a path exists in only 30–32% of these deliberately hard tasks, against 98–100% on the true map. Fig. 1 shows a typical case. On random tasks the joint planner succeeds in 78% of the missions without drift and in 68% at L3, with no unsafe mission; the uncalibrated one succeeds in 95% and 82%, but 4–6% of its missions are unsafe. Map quality does not predict safety: across scenes, the Spearman correlation between mIoU and the violation rate of the uncalibrated planner is -0.16 ($p = 0.57$, 15 scenes).

C. Comparison with per-cell label calibration in its setting

The closest prior work [6] assumes a known pose and a closed vocabulary and reports results on random tasks. We reproduce these conditions on our scenes: known pose, a closed vocabulary of five classes (five rows of the detector’s prompt embeddings, which equals re-prompting the detector), $1 - \alpha = 0.95$ and random tasks; their exploration mode is not reproduced. Fig. 2 and Table II show the result. In their setting the two calibrations are equal (0.43 vs. 0.42 mission success, no unsafe missions). Once the pose drifts, the coverage of label sets falls to 0.42 at 2 cm and 0.10 at 7 cm, while the joint margin stays at 0.94 and succeeds more often (0.61 and 0.50 vs. 0.40 and 0.32). With the open vocabulary the joint margin succeeds 1.8× more often already at known pose (0.78 vs. 0.43).

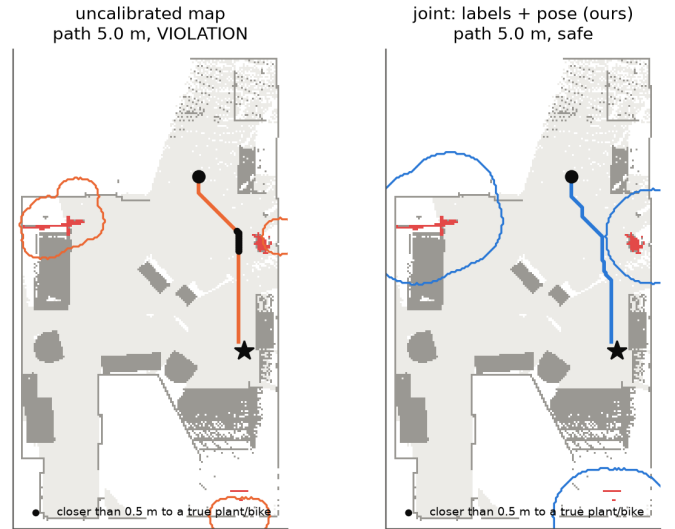


Fig. 1. A pass-by task at L2 (2 cm ATE). Red: true plant and bike; line: keep-out zone. The uncalibrated zone misses part of the plant and the shortest path passes within 0.5 m of it (black dots); the joint zone covers it.

TABLE II
RANDOM TASKS: COVERAGE (MINIMUM OVER PLANT AND BIKE), MISSION SUCCESS AND UNSAFE MISSIONS.

setting	drift	per-cell labels [6]			joint (ours)		
		cov.	succ.	unsafe	cov.	succ.	unsafe
closed vocab., $1 - \alpha = 0.95$	known	1.00	0.43	0.000	0.94	0.42	0.000
	2 cm	0.42	0.40	0.000	0.94	0.61	0.000
	7 cm	0.10	0.32	0.002	0.94	0.50	0.000
open vocab., $1 - \alpha = 0.9$	known	1.00	0.43	0.000	0.93	0.78	0.000
	2 cm	0.41	0.40	0.000	0.93	0.74	0.000
	7 cm	0.10	0.32	0.002	0.94	0.68	0.000

Why do label sets degenerate here? We first suspected the fine grid and repeated both calibrations at known pose on grids of 5, 25, 50 and 100 cm. The calibrated threshold is $\lambda^* = 0$ at every resolution, and the keep-out zone of label sets removes 64% of the free space at 5 cm and 100% at 1 m. The cause is complete misses: in some scenes no observation of the class reaches part of the object, so the worst-cell score is 1 and the threshold is zero. A label set can only say “this cell may be a plant”; a distance can say “a plant is nearby”, which remains informative when the detector misses parts of objects.

D. The price of the guarantee

Where does the remaining gap to the oracle come from? Raising the risk level does not help much: from $\alpha = 0.1$ to 0.3 the joint margin shrinks from 0.81 to 0.60 m (plant) and from 0.75 to 0.50 m (bike), and pass-by tasks with a path go from 30% to 31% (L3), while the joint coverage tracks $1 - \alpha$ at every level (0.94, 0.86–0.87, 0.72). Better masks do not help either: MobileSAM [18] prompted with the detector boxes cuts the median bike miss from 0.43 to 0.05 m, but the worst scene keeps a 0.75 m miss and, with 13 calibration scenes, sets the margin. With SAM masks the uncalibrated planner becomes

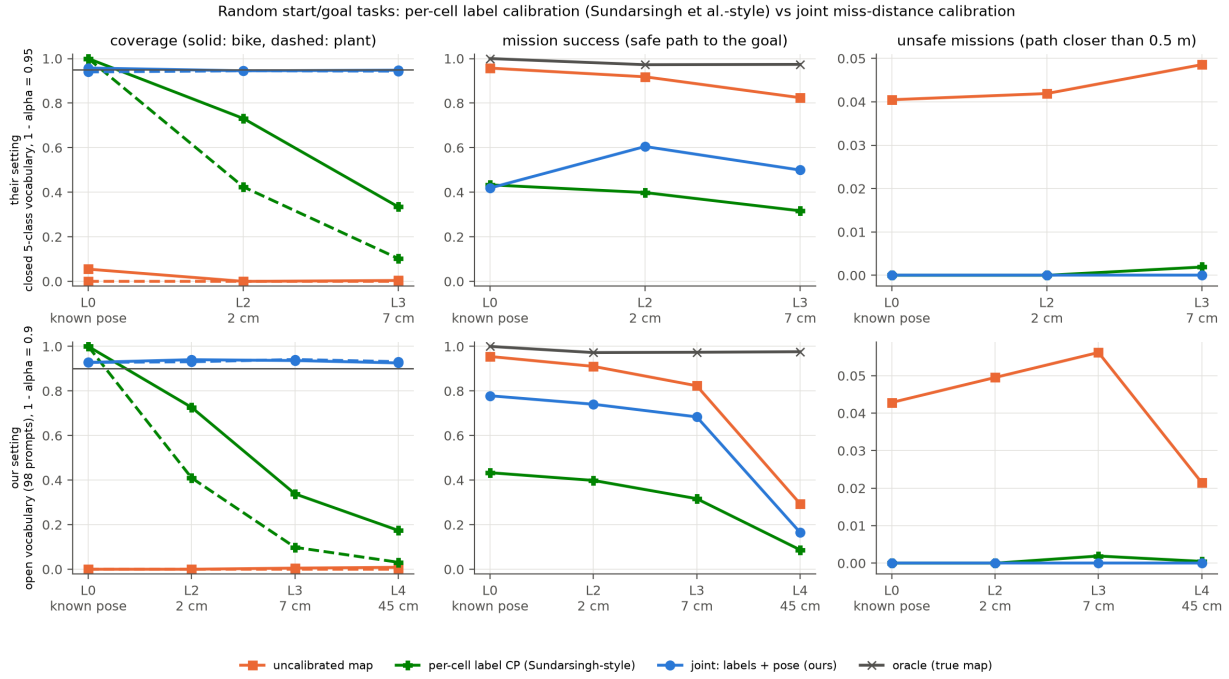


Fig. 2. Per-cell label-set calibration [6] vs. joint miss-distance calibration on random tasks, in the setting of [6] (top: closed vocabulary, $1 - \alpha = 0.95$) and ours (bottom: open vocabulary, $1 - \alpha = 0.9$).

safe at known pose (no unsafe paths) but not under drift (9.5% unsafe paths at L3). The margin is set by the tail of perception errors; reducing it requires more calibration scenes or better worst-case perception, not a bolder risk level.

E. When does the sum of quantiles fail?

Synthetic drift is independent of detector errors, and there the sum of separate quantiles over-pays (Table I). To probe dependence, we keep the real label errors and replace the drift by a scene-level mixture: in k of 21 scenes a localization failure occurs (realizations from L5), in the rest drift is negligible (L1). Failures are placed in the scenes with the worst labels (aligned), the best labels (disjoint) or at random. At $\alpha = 0.3$ (at $\alpha = 0.1$ with 13 scenes rare failures always enter the maximum) the sum covers 0.76–0.84 when aligned but only 0.66 at a 0.70 target when disjoint (plant, 19–24% failing scenes); the joint quantile stays at 0.71–0.78 in every configuration (Fig. 3).

F. Real odometry and other scenes

Frame-to-frame RGB-D odometry without loop closure gives a heavy-tailed error across scenes: ATE from 3.5 cm to 3.9 m, median 78 cm, mostly from single failures during in-place turns near walls. The labels-only margin then silently covers only 0.67 (bike); the joint margin stays valid (0.93) but abstains in 60% (bike) and 86% (plant) of the splits. In missions the uncalibrated planner finds a path in 23% of the tasks and 44% of these paths are unsafe; the joint arm abstains in 89% of the tasks. Without loop closures there is no useful certificate, and the calibration reports it instead of

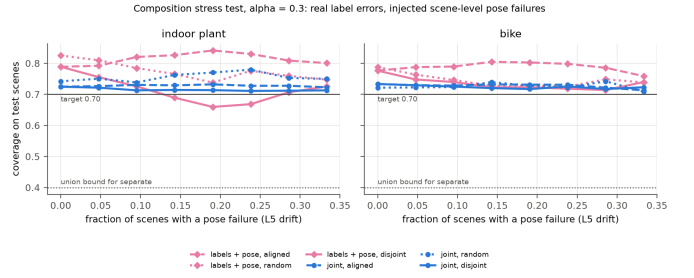


Fig. 3. Coverage of separate (pink) and joint (blue) calibration vs. the fraction of scenes with a localization failure, placed where labels are worst (dashed), best (solid) or at random (dotted); $1 - \alpha = 0.7$.

issuing a false one. Calibrated on ReplicaCAD and tested on HM3D scenes [19] of OSMA-Bench, the guarantee does not transfer: of the two scenes in which a plant is visible from the trajectory, the detector finds one (miss 0.18 m) and misses the other (2.9 m).

G. Planner on calibrated maps

Calibration narrows the free space. We compare RRT-Connect, the angle-limited dynamic sampling zone of a bidirectional RRT [20] (cone 8° , step 2° , widened while samples hit obstacles) and a variant that widens the cone after blocked tree extensions, on pass-by tasks at L2 with a 1500-extension budget and the margins used in the missions. With the calibrated margin only 30% of the tasks are solvable (97% without margin); solvable tasks are solved within the budget about equally often by all three (0.86–0.89). The

published rule needs 2.3–3.0× more extensions than uniform sampling, because on tightened maps the passages are offset from the line between the trees and few samples fall into obstacles. Widening on blocked extensions brings this to 1.2–1.4× without longer first paths (1.11–1.17 of optimal vs. 1.15–1.21 for uniform sampling).

VII. DISCUSSION AND LIMITATIONS

The joint margin is not better than label-set calibration where the latter’s assumptions hold (known pose, closed vocabulary); it is better exactly where they fail. The bottleneck is not drift or boundary errors but confident misses and class swaps of whole objects, which no class-specific margin can certify at a risk level below their frequency. Map metrics such as mIoU average over cells and do not reflect it; a “fraction of scenes with a whole-object miss” metric would be a natural addition to mapping benchmarks.

Limitations: one family of 21 simulated scenes and one lighting condition (the other OSMa-Bench conditions are not published); a box detector rather than a full mapper; the main drift is synthetic, and real odometry was tested only without loop closure; the guarantee is marginal over scenes and holds in the planner frame at query time, not during execution; the out-of-distribution check covers two scenes. The choice of plant and bike as mission classes was made after seeing that the TV stand abstains; both configurations are reported in the code repository.

VIII. CONCLUSION

One calibrated quantity, the miss distance in the planner frame, gives a planner a distribution-free guarantee against label and pose errors together. Margins calibrated on either error alone lose coverage under the other, and the sum of separate margins depends on an unknown error dependence. In the setting of the closest prior work our calibration matches it; with drift or an open vocabulary it keeps the guarantee and succeeds more often. The price is set by the worst perception errors. Next steps are SLAM with loop closures, more calibration scenes, object-dependent margins, and planners that operate efficiently in the narrowed free space.

REFERENCES

- [1] Q. Gu *et al.*, “ConceptGraphs: Open-vocabulary 3D scene graphs for perception and planning,” in *Proc. IEEE ICRA*, 2024, pp. 5021–5028, doi: 10.1109/ICRA57147.2024.10610243.
- [2] M. Popov *et al.*, “OSMa-Bench: Evaluating open semantic mapping under varying lighting conditions,” in *Proc. IEEE/RSJ IROS*, 2025, pp. 10786–10791, doi: 10.1109/IROS60139.2025.11247603.
- [3] R. Kurkova, M. Popov, and S. Kolyubin, “OSMa-Bench++: Toward open-ended benchmarking of semantic mapping for manipulation with prompt-generated synthetic scenes,” arXiv:2605.26831, 2026 (ICRA 2026 workshop).
- [4] D. Rotondi *et al.*, “3D scene graphs: Open challenges and future directions,” arXiv:2606.19383, 2026.
- [5] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” arXiv:2107.07511, 2021.
- [6] D. S. Sundarsingh *et al.*, “Safe planning in unknown environments using conformalized semantic maps,” *IEEE Robot. Autom. Lett.*, vol. 11, no. 4, pp. 4921–4928, 2026, doi: 10.1109/LRA.2026.3668466.
- [7] Z. Mei *et al.*, “Perceive with confidence: Statistical safety assurances for navigation with learning-based perception,” *Int. J. Robot. Res.*, vol. 45, no. 6, pp. 938–967, 2025, doi: 10.1177/02783649251378151.
- [8] H. Sier *et al.*, “P-POSEMEM: Projective semantic memory for consistent language grounding under pose-graph rewrites,” arXiv:2609.15475, 2026.
- [9] R. Baral, “Marginal calibration does not compose: Hidden dependence in modular robot navigation,” arXiv:2609.23731, 2026 (IROS 2026 workshop).
- [10] J. M. Correia Marques, A. Zhai, S. Wang, and K. Hauser, “On the overconfidence problem in semantic 3D mapping,” in *Proc. IEEE ICRA*, 2024, pp. 11095–11102, doi: 10.1109/ICRA57147.2024.10611306.
- [11] L. Lindemann *et al.*, “Safe planning in dynamic environments using conformal prediction,” *IEEE Robot. Autom. Lett.*, vol. 8, no. 8, pp. 5116–5123, 2023, doi: 10.1109/LRA.2023.3292071.
- [12] D. Kumar *et al.*, “Learnable conformal prediction with context-aware nonconformity functions for robotic planning and perception,” in *Proc. IEEE ICRA*, 2026, arXiv:2509.21955.
- [13] S. Natraj, B. Sinopoli, and Y. Kantaros, “Conformal constraint tightening for chance-constrained motion planning with unknown dynamics,” arXiv:2607.22409, 2026.
- [14] A. Szot *et al.*, “Habitat 2.0: Training home assistants to rearrange their habitat,” in *Proc. NeurIPS*, 2021, arXiv:2106.14405.
- [15] T. Cheng *et al.*, “YOLO-World: Real-time open-vocabulary object detection,” in *Proc. IEEE/CVF CVPR*, 2024, pp. 16901–16911, doi: 10.1109/CVPR52733.2024.01599.
- [16] Q.-Y. Zhou, J. Park, and V. Koltun, “Open3D: A modern library for 3D data processing,” arXiv:1801.09847, 2018.
- [17] F. Steinbrücker, J. Sturm, and D. Cremers, “Real-time visual odometry from dense RGB-D images,” in *Proc. IEEE ICCV Workshops*, 2011, pp. 719–722, doi: 10.1109/ICCVW.2011.6130321.
- [18] C. Zhang *et al.*, “Faster segment anything: Towards lightweight SAM for mobile applications,” arXiv:2306.14289, 2023.
- [19] S. K. Ramakrishnan *et al.*, “Habitat-Matterport 3D dataset (HM3D): 1000 large-scale 3D environments for embodied AI,” in *NeurIPS Datasets and Benchmarks*, 2021, arXiv:2109.08238.
- [20] I. S. Dovgopoli and O. I. Borisov, “Quasi-optimal shortest-path motion planning algorithm with random sampling for robotic systems,” *Izv. Vyssh. Uchebn. Zaved., Priborostroenie*, vol. 68, no. 5, pp. 450–455, 2025, doi: 10.17586/0021-3454-2025-68-5-450-455 (in Russian; title translated).